

RE02 Real Estate Finance and Investment

Michaelmas Term 2026, Week 2

Space and Capital Markets, Pricing Processes

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What this week does

Real estate is traded in two markets at once. Occupiers buy the use of space and pay rent for it; investors buy the income that rent produces and pay a capital price for it. The two markets have different participants, different clearing speeds and different information, yet they price the same buildings. This week sets out the standard framework for holding them together, and spends the first half on the reason the framework is needed, which is that the adjustment between the two is slow enough for prices to move a long way from anything a long-run equilibrium would support.

A	Real estate and the macroeconomy	25 min
B	Construction, talent and the allocation of resources	25 min
C	The four quadrants	30 min
D	Comparative statics in the four quadrants	35 min
E	The quadrants as a system of equations	25 min
F	Cambridge in 2022, and what happened next	35 min
G	What the model leaves out	25 min
H	Reserve: where the model is used next	20 min

What you already have

Where this week draws on material taught elsewhere in RE02. Nothing below is assumed: each idea is stated in full here before the other lecture is named.

Where it was taught	What it gives this week	What happens if it was not covered
RE02 Week 1, introduction and macroeconomic analysis	The size and composition of the commercial stock, the inelasticity of supply in the short run, and the main macroeconomic drivers of the sector.	Block A restates the inelastic-supply point in full before it is used. Nothing later in the week depends on Week 1 having been delivered in any particular form.
Undergraduate microeconomics, any course	Comparative statics: the effect of shifting one schedule on the equilibrium of a system.	Block D works the first shift slowly and names each step. No prior facility with the technique is assumed.
Where this week is used again	The capitalisation ray in the north-west quadrant is the yield. The stock adjustment quadrant is the supply pipeline. Both reappear in the valuation lectures and in Week 8.	This lecture supplies those ideas rather than borrowing them, so it can be delivered in any position in the term.

Block A

Real estate and the macroeconomy

Why a lecture on pricing has to begin with credit.

Indicative length 25 minutes

The asset that prices two things at once

A commercial building generates a stream of rent, and it is also a claim that can be bought and sold. Those are separate markets with separate participants. An occupier compares the rent against the cost of doing without the space; an investor compares the yield against what a bond or an equity would pay. The building is the same object in both cases, which means the two markets cannot price it independently for long.

What makes real estate distinctive is the length of time they can disagree. Supply cannot respond within a year, existing leases fix the income for a decade, and the asset is expensive enough that most transactions are financed. The result is that the adjustment from one long-run position to another runs through a sequence of short-run positions in which price is a poor guide to value.

Four features that slow the adjustment

Durability

A building put up in a boom is still standing three cycles later. Supply cannot be withdrawn when demand falls, so the correction falls entirely on price and on vacancy rather than on quantity.

Construction lags

Two to four years between the decision to build and delivery, longer where consent is contested. Starts respond to the price that prevailed when the decision was taken, not the price at completion.

Immobility

Space cannot be moved to where the demand is. A national vacancy rate conceals shortage in one city and surplus in another, and arbitrage between them is impossible.

Credit intensity

Most acquisitions are financed. The price the asset market will pay therefore depends on the terms on which debt is available, which are set outside the sector entirely.

Disequilibrium is the normal condition, not the exception

It is tempting to treat episodes of severe mispricing as departures from a market that otherwise clears. The sequence set out above suggests the opposite. Given lags of several years and a stock that cannot shrink, a market hit by a demand shock will spend most of the intervening period away from any long-run equilibrium, and will only pass through the equilibrium on its way somewhere else.

This has a practical consequence for the rest of the module. A valuation that assumes the market is at rest will be wrong in a predictable direction, and knowing which direction requires knowing where in the cycle the market currently sits.

When credit and prices reinforce each other

Rising collateral values relax the constraint on borrowing, and relaxed borrowing raises what buyers can pay, which raises collateral values again. The loop is familiar and it is the reason real estate booms are usually credit booms with a real estate label. Jordà, Schularick and Taylor examine the point across a long panel of advanced economies and find that the busts that follow leveraged booms are markedly more damaging than those that follow booms financed from equity.

Zhao and colleagues put the same question to the Chinese housing market and reach a more qualified answer, arguing that the conditions usually taken to signal an imminent correction had been present for years without one occurring. The disagreement is instructive: the mechanism is not in dispute, the timing is.

What the macroprudential toolkit can and cannot do

- The instruments divide into those aimed at the borrower, principally limits on loan-to-value and on debt service relative to income, and those aimed at the lender, principally capital and provisioning requirements.
- Kuttner and Shim examine both across a panel of fifty-seven economies and find the borrower-side instruments the more reliable, with limits on debt service relative to income the most consistently effective.
- The effect on credit growth is easier to detect than the effect on prices. This is worth dwelling on, because the political case for these tools is usually made in terms of prices.
- None of the instruments addresses the supply side, so a market whose problem is a binding constraint on new building will not be fixed by any of them.
- The interest rate remains the blunt instrument. It moves the whole economy in order to reach one sector, which is the reason the macroprudential toolkit was developed in the first place.

Block A discussion

Seminar questions

1. If supply cannot respond for four years, what does a rent quoted today actually tell an investor about the rent that will be received in year five?
2. A regulator can restrict how much a household may borrow, or it can restrict how much a bank may lend. Which produces the more predictable effect on prices, and why do the two differ?
3. The evidence that leveraged booms are followed by worse busts is well established. Why has it proved so hard to act on in advance?

Block B

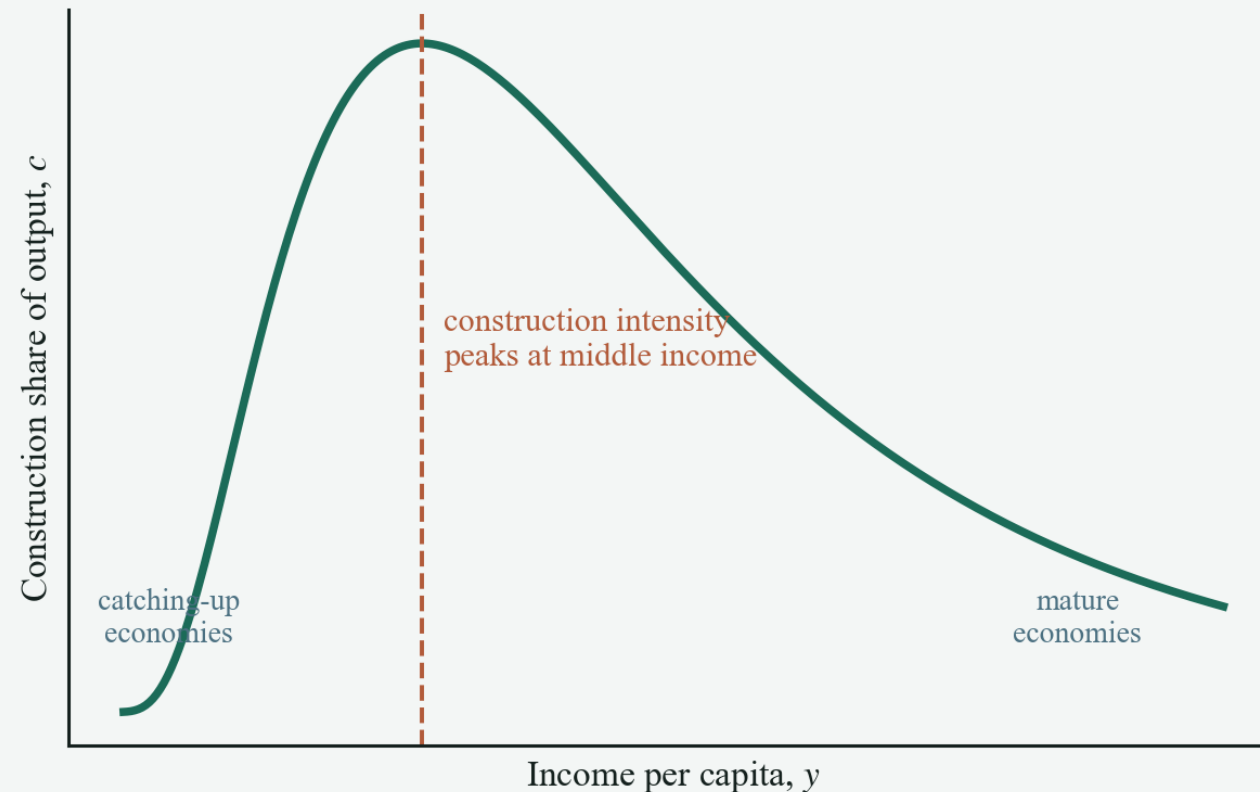
Construction, talent and the allocation of resources

What a large real estate sector does to the economy that hosts it.

Indicative length 25 minutes

Construction intensity rises and then falls with income

Construction absorbs a rising share of output while an economy is building its stock, and a falling share once the stock is largely in place and the work shifts towards maintenance and replacement.



What the inverted U implies for a development pipeline

The curve is a statement about composition rather than about volume. In a catching-up economy the pipeline is dominated by additions to the stock, and the binding constraint is usually finance or construction capacity. In a mature economy the pipeline is dominated by replacement and refurbishment, and the binding constraint is usually consent and the existing building itself.

That difference changes what a forecast has to model. Where new supply dominates, the forecaster needs a starts equation. Where replacement dominates, the forecaster needs a depreciation and obsolescence rate, which is a much harder thing to observe and a much less stable parameter.

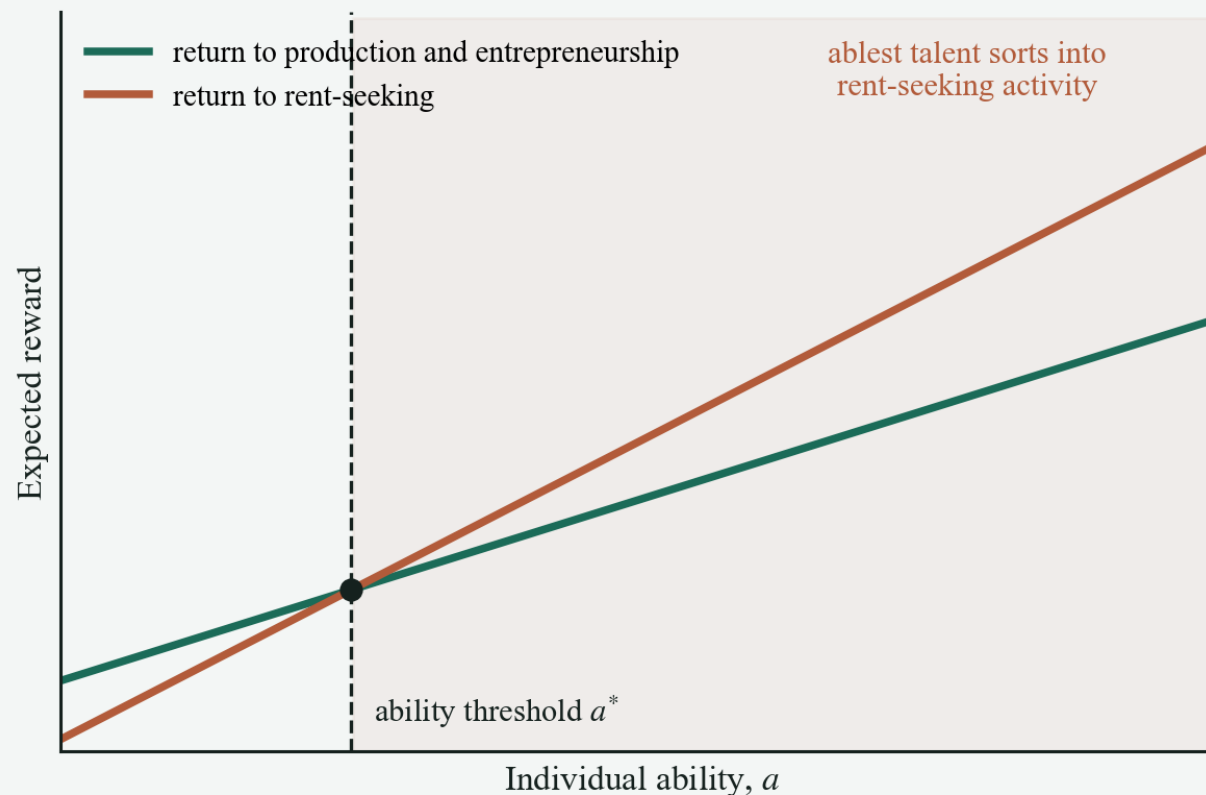
Entry competes away the rent, but not in the way one might hope

Hsieh and Moretti study residential brokerage in the United States and find a pattern that is easy to state and uncomfortable to think about. Where land prices rise, the number of brokers rises with them, because commission is charged as a percentage of a price that is now higher. Entry continues until the return to brokerage is back in line with the alternatives. The result is that the average broker sells fewer houses per hour worked and earns much the same real wage as before.

The rent created by rising prices is therefore dissipated rather than captured. It is spent on the additional brokers required to compete for a fixed number of transactions.

The allocation of ability between production and rent-seeking

Where the reward to rent-seeking rises more steeply with ability than the reward to production does, the ablest people sort into the activity that redistributes existing wealth rather than the one that creates new wealth.



Schematic, after Murphy, Shleifer and Vishny (1991).

Why this belongs in a real estate module

Murphy, Shleifer and Vishny are not writing about property. Their argument is general: an economy in which the highest returns accrue to activities that capture value rather than create it will grow more slowly than its endowment of talent would predict, because the talent is in the wrong place.

Real estate is a natural candidate for such an activity, since a great deal of the return to landholding is a transfer from the users of land rather than a payment for anything produced. Whether a given market is closer to the productive case or the rent-seeking one depends on how far returns come from development and improvement, and how far from planning permission and scarcity. That is an empirical question, and it is one worth asking of any market a student is about to write about.

London as a case in point

- The share of London output accounted for by real estate and by activities ancillary to it has risen substantially over three decades, faster than in comparable European cities.
- A rising share is not by itself evidence of misallocation. It is consistent with London genuinely specialising in a service it provides well and exports.
- The diagnostic question is whether the return comes from producing space or from holding it. Development, refurbishment and management are production; capturing the uplift on a consent is not.
- The distinction is hard to make in the national accounts, which is one reason the argument has not been settled empirically.
- It matters for policy because the two cases call for opposite responses, one of encouragement and one of taxation.

Block B discussion

Seminar questions

1. Brokerage commission is charged as a percentage of price. What would change if it were charged as a fixed fee per transaction?
2. How would you tell, from data you could actually obtain, whether returns in a given city came mainly from producing space or mainly from holding it?
3. If the ablest graduates of this department go into real estate, is that evidence for the Murphy, Shleifer and Vishny argument, or against it?

Block C

The four quadrants

The standard framework for holding the space market and the asset market together.

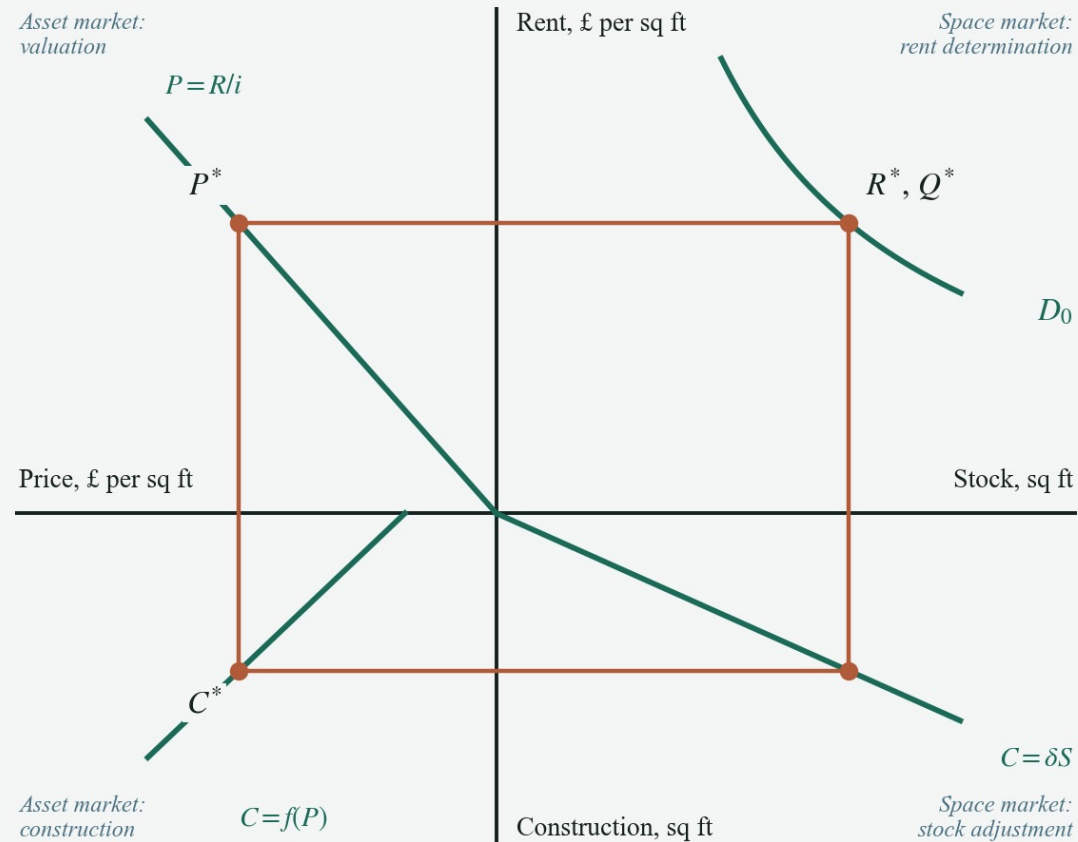
Indicative length 30 minutes

Two markets, one building

	Space market	Asset market
What is traded	The use of space for a period	The ownership of the income stream
Who trades	Occupiers and landlords	Investors, funds and lenders
The price	Rent per square foot per year	Capital value per square foot
What determines it	Employment, output and the stock of space	The yield available on competing assets
Speed of adjustment	Slow, and slowed further by lease length	Fast, and continuous
Link to the other	Rent is the income the asset market capitalises	Price is the signal that triggers construction

The four quadrants in long-run equilibrium

Reading anticlockwise from the north-east, rent is determined in the space market, capitalised into a price, the price calls forth construction, and construction replaces the stock that rent was determined on. Equilibrium requires that the path returns to where it began.



What each quadrant does

North-east: rent determination

The demand schedule for space. Given the stock in existence and the level of employment, this fixes the rent at which the space market clears. Its slope is the rental elasticity of demand.

North-west: valuation

The capitalisation ray, P equals R divided by i . Its slope is the capitalisation rate, which the model takes as given from the wider capital market rather than determining.

South-west: construction

The supply of new building as a function of price. It starts at a threshold price below which nothing is built, and its slope is the price elasticity of construction.

South-east: stock adjustment

The steady-state condition. Construction must replace what depreciates each year, so C equals δ times S , where δ is the rate at which the stock is lost.

Why the rectangle has to close

Take any rent in the north-east quadrant and follow it round. It implies a price, the price implies a rate of construction, and that rate of construction implies a stock. If the stock the path produces is not the stock the rent was read off, the starting point was not an equilibrium and the system will move.

The closure requirement is what makes the diagram a model rather than a set of graphs. It also makes the comparative statics tractable: shift any one schedule and the new equilibrium is the unique rectangle that closes against the shifted schedule and the three unchanged ones.

Block C discussion

Seminar questions

1. The capitalisation rate enters the model from outside. What would have to be true of the world for that to be a reasonable assumption?
2. The stock adjustment quadrant assumes a constant depreciation rate. Which property types would you expect to violate that most badly?
3. Nothing in the diagram represents vacancy. Where would you put it, and what would it change?

Block D

Comparative statics in the four quadrants

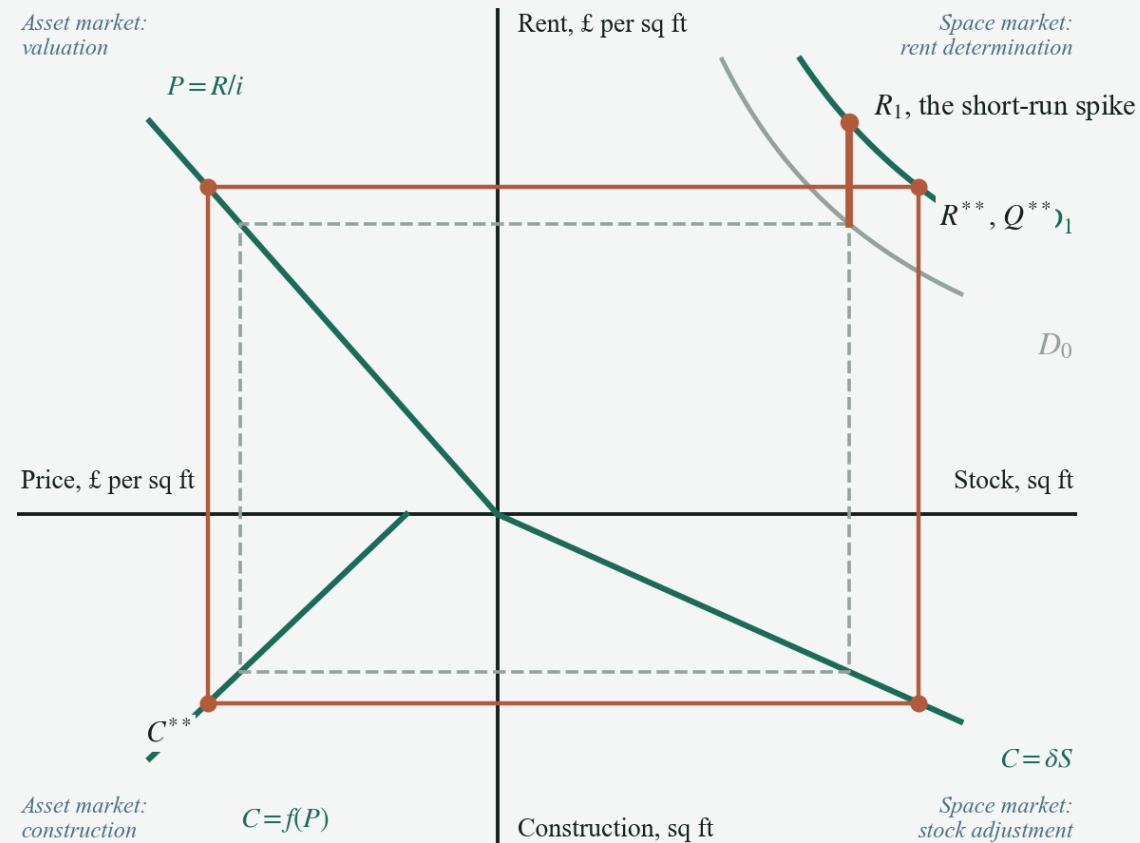
Two shocks, worked through slowly.

Indicative length 35 minutes

Demand growth in the space market

D

Employment rises, so the demand schedule for space shifts out. The stock cannot change immediately, so rent must absorb the whole of the shock, and only later does new building take the pressure off.



After diPasquale and Wheaton (1992).

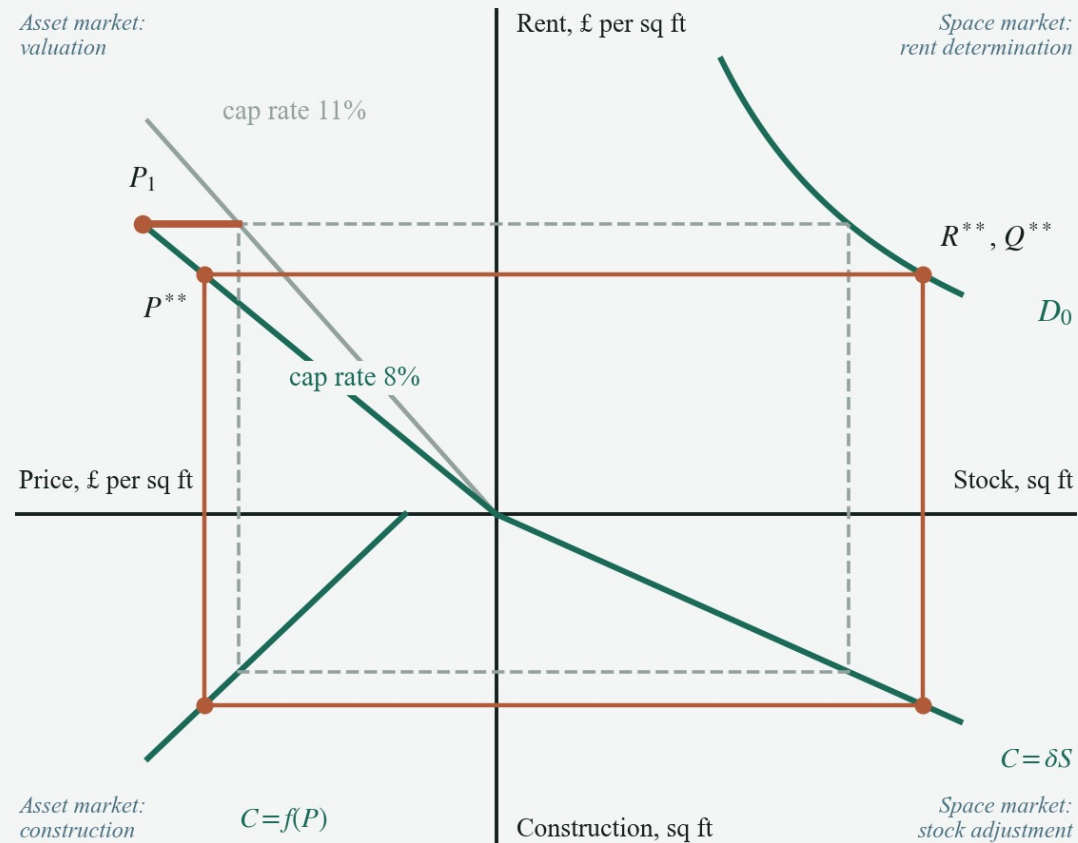
The rent path after a demand shock

- On impact the stock is fixed, so the whole of the additional demand is met by a higher rent. This is the spike marked on the diagram, and it is the highest rent the market will see.
- The asset market capitalises the higher rent immediately, so price rises at once and by more than the eventual long-run increase.
- The higher price makes development viable, and starts rise. Nothing arrives yet, because of the construction lag.
- Completions raise the stock. The demand schedule is unchanged, so a larger stock clears at a lower rent than the spike.
- The new long-run rent lies above the old one but below the spike. New supply tempers the initial rise without reversing it.
- The size of the overshoot depends on the price elasticity of construction. An inelastic supply response gives a higher spike and a longer period at elevated rents.

Yield compression in the asset market

D

The capitalisation rate falls, so the valuation ray flattens and the same rent supports a higher price. The shock arrives from the capital market rather than from occupiers, and its effect on rent runs in the opposite direction.



After diPasquale and Wheaton (1992).

Why cheaper capital lowers rents

A fall in the capitalisation rate raises price without touching the demand for space. The higher price makes development viable at the margin, so construction rises and the stock grows. A larger stock, facing an unchanged demand schedule, clears at a lower rent.

The result is worth stating plainly because it runs against intuition. A sustained fall in yields is good for existing owners on the day it happens and bad for their rental income thereafter. It also explains a pattern that is otherwise puzzling, in which capital values and rental values move in opposite directions for years at a time.

Four shocks and the direction of the response

Shock	Rent	Price	Construction	Stock
Employment rises	Up	Up	Up	Up
Capitalisation rate falls	Down	Up	Up	Up
Construction costs rise	Up	Up	Down at first, then to the new steady state	Down
Depreciation rate rises	Up	Up	Up	Down

Block D discussion

Seminar questions

1. A market has a strict consent regime and therefore a low price elasticity of construction. Trace a demand shock through the diagram and say how the rent path differs from an unconstrained market.
2. Yields fall by two hundred basis points and rents are unchanged for three years. Is the market in equilibrium, and if not, what has to happen next?
3. Which of the four schedules would you expect to have shifted most in the London office market since 2020, and what does the model predict should follow?

Block E

The quadrants as a system of equations

The same model, written down.

Indicative length 30 minutes

The vocabulary the algebra requires

- Parameters are constants that describe behaviour and are treated as fixed within the exercise. In this model they are the elasticities and the depreciation rate.
- Endogenous variables are the ones the model determines. Here they are construction, stock, rent and price, which are exactly the four axes.
- Exogenous variables are determined outside the model and are not affected by it. Here they are employment and the capitalisation rate.
- An equilibrium is a set of values for the endogenous variables that satisfies every equation simultaneously, given the parameters and the exogenous variables.
- Comparative statics is the exercise of changing one exogenous variable and comparing the new equilibrium with the old, which is precisely what Block D did with a pointer.

The first quadrant: demand for space

Write demand for office space as a constant multiplied by employment and by rent raised to a negative power. Employment scales the requirement, and the exponent is the rental elasticity of demand, the proportional change in space taken per worker for a given proportional change in rent.

In equilibrium the space demanded equals the stock in existence, so the relationship can be inverted to give rent as a function of the stock and of employment. That inverted form is the downward-sloping schedule in the north-east quadrant. The constant has a simple interpretation: it is the space taken per worker when rent is one pound.

The second and third quadrants: valuation and construction

Valuation is the simplest relationship in the model. Price is rent divided by an all-inclusive capitalisation rate, which is taken as given. The word inclusive is doing work: the rate has to absorb the expected growth in rent, the risk of the income and the cost of capital, all in one number.

Construction is written as a rate rather than a level, with the ratio of new building to existing stock rising with price. The exponent is the price elasticity of supply. Writing it as a ratio is convenient rather than innocent, because it implies that a market with twice the stock will build twice as much in response to the same price, which is not obviously true of markets constrained by land rather than by capital.

The fourth quadrant: two ways to close the system

The stock adjustment quadrant can be closed in either of two ways, and they answer different questions. In the replacement version, employment is held fixed and the stock is eroded at a constant rate, so equilibrium requires that construction exactly offsets the loss. This is the version drawn in the diagram, and it describes a mature market.

In the steady-growth version, employment grows at a constant rate and there is no physical loss, so equilibrium requires that the stock grow at the same rate as employment. The interesting case is the one in between, in which the ratio of construction to stock differs from the sum of the depreciation and growth rates, and space per worker is therefore changing.

The system in one place

Quadrant	Relationship	Parameter	What it measures
North-east	Rent falls with the stock, given employment	Rental elasticity of demand	How far occupiers economise on space when rent rises
North-west	Price is rent divided by the capitalisation rate	Capitalisation rate	The yield the capital market requires, taken as given
South-west	Construction rises with price above a threshold	Price elasticity of supply	How readily the development industry responds
South-east	Construction replaces what is lost	Depreciation rate	How fast the existing stock ceases to be useful

Block E discussion

Seminar questions

1. The capitalisation rate is exogenous and the depreciation rate is a parameter. Which of the two would you find harder to estimate for a real market, and why?
2. Space per worker in central London offices has fallen over three decades. Which schedule does that represent, and is it a movement along it or a shift of it?
3. The construction equation is written as a ratio to existing stock. Construct a case in which that formulation gives the wrong answer.

Block F

Cambridge in 2022, and what happened next

The same four quadrants, at observed values, tested against the outturn.

Indicative length 35 minutes

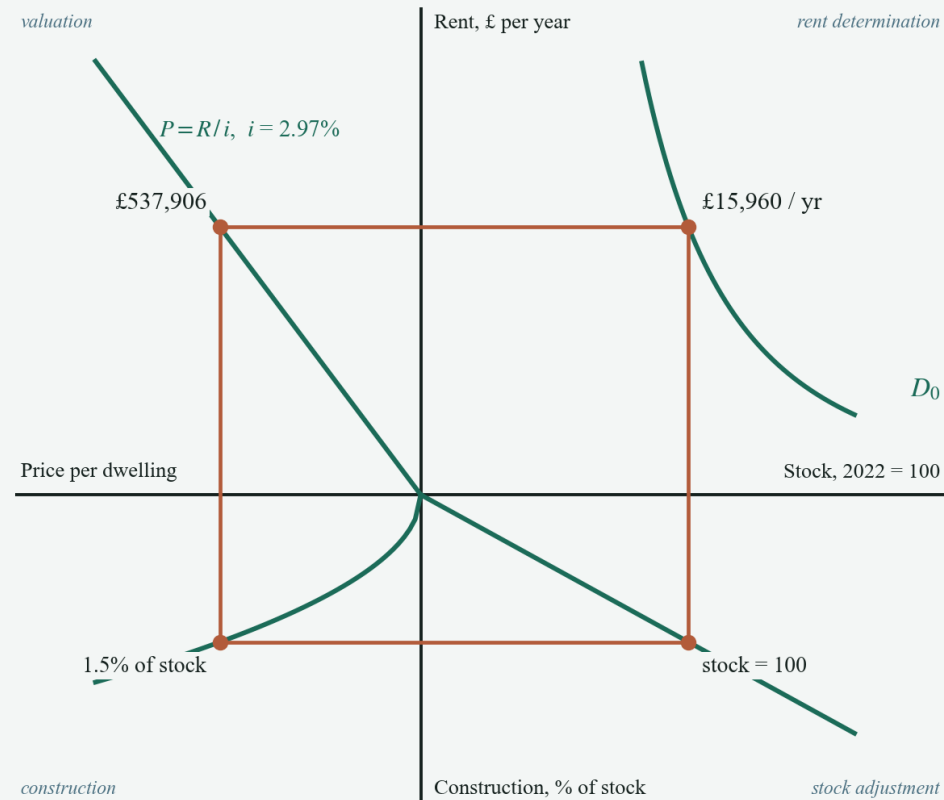
From a diagram to a market

Everything so far has been drawn without units. The curves had shapes and the shifts had directions, but nothing on either axis was a number anyone could look up. That is the right way to learn the mechanics and the wrong way to find out whether they are any use.

So take one market. Cambridge in 2022 had an average advertised rent of £1,330 a month and an average sold price of £537,906, both observed rather than assumed. Two of the four quadrants can be pinned to those numbers immediately. The other two cannot, and seeing exactly where the data stops and judgement starts is half the value of the exercise.

The four quadrants, Cambridge 2022

The rent, the price and the ray connecting them are observed. The stock is set to 100 in the base year and construction to the replacement rate, because neither is in the data.



Rent and price observed; stock and construction indexed.

What is observed, and what is not

Quantity	Cambridge 2022	Where it comes from
Rent	£1,330 per month, £15,960 per year	Geometric mean advertised rent, observed
Price	£537,906	Geometric mean sold price, observed
Capitalisation rate	2.97 per cent gross	The ratio of the two, so also observed
Stock	Indexed to 100	Not in the data. The index is a normalisation, not a count of dwellings
Construction	Set to the replacement rate	Not in the data. It follows from the replacement rate you choose

The system, written at index values with 2022 as one

Quadrant	Equation	What it says
North-east, rent determination	$R = g^{(1/\epsilon)} \cdot S^{(-1/\epsilon)}$	Rent falls as the stock grows, at a rate governed by ϵ
North-west, valuation	$P = (i_0 / i_1) \cdot R$	Price is rent capitalised at the yield, relative to the observed 2.97 per cent
South-west, construction	$S = (P / k)^{\beta}$	Building, and so the stock it sustains, rises with price and falls with cost
South-east, stock adjustment	$C = \delta \cdot S$	In steady state, building replaces what is lost each year
Solving the four together	$P = [(i_0/i_1) \cdot g^{(1/\epsilon)} \cdot k^{(\beta/\epsilon)}]^{(\epsilon/(\epsilon+\beta))}$	The one price at which the rectangle closes; R and S follow from it
On impact, stock fixed at one	$R_1 = g^{(1/\epsilon)}, P_1 = (i_0/i_1) \cdot g^{(1/\epsilon)}$	The short-run position before any building responds

Money values are the index multiplied by £15,960 for rent and £537,906 for price.

Which control changes which symbol

Control	Box	Symbol	Enters	What moves on the diagram
Rental elasticity of demand	2	ϵ	Equation 1	The steepness of the north-east curve
Price elasticity of new supply	2	β	Equation 3	The steepness of the south-west curve
Replacement rate	2	δ	Equation 4	The construction axis only. Rent, price and stock are untouched
Demand for space	3	g	Equation 1	The north-east curve shifts out or in
Capitalisation rate	3	i_l	Equation 2	The north-west ray pivots
Construction cost	3	k	Equation 3	The south-west curve shifts

Why the replacement rate moves nothing but construction

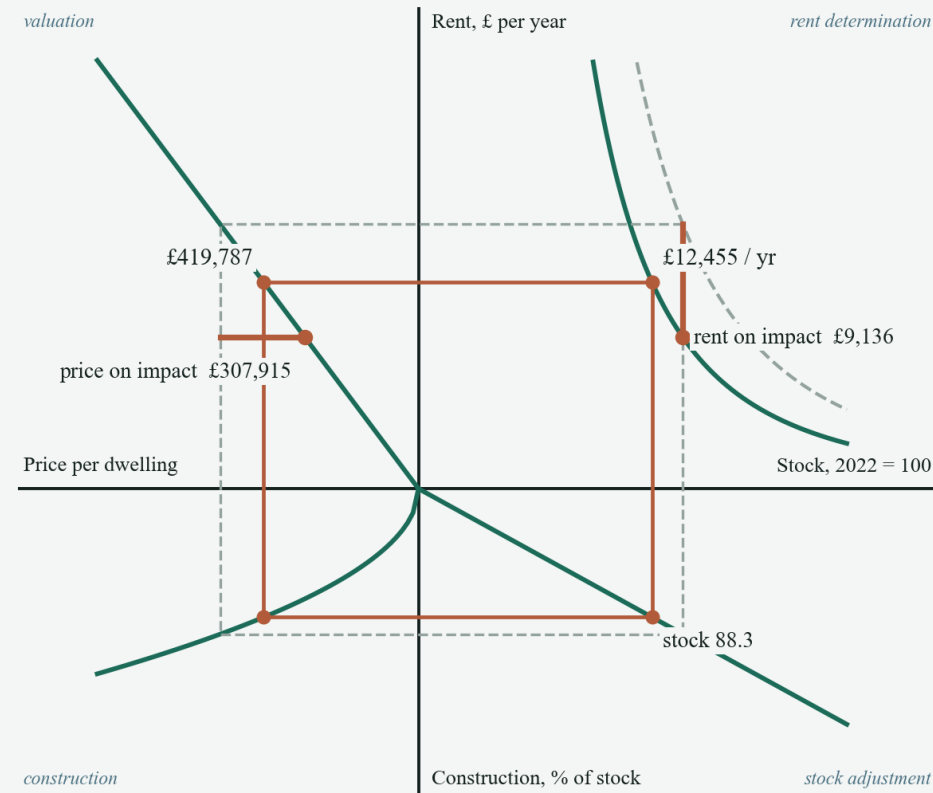
One control behaves differently from the rest and it is worth pausing on, because a student who moves it and sees nothing happen will assume the tool is broken. In steady state, building replaces what is lost, so construction and the stock it supports move in exact proportion. Working in index numbers, that proportion cancels: doubling the replacement rate doubles both the building required and the building delivered, and leaves rent, price and the stock index exactly where they were.

The rate is still doing work. It converts the stock index into a rate of building per year, which is what a development pipeline is measured in. It simply has no bearing on where the equilibrium sits, and the fact that a parameter can matter for one purpose and not another is worth more than the arithmetic that shows it.

Scenario one: the search data, taken at face value

F

Search enquiries for Cambridge fell by about twenty per cent between 2021 and 2023. Entered as a fall in the demand for space, the model says rent should collapse on impact to £9,136 and settle at £12,455, with price falling by the same proportion.



Demand shift from search enquiries; $\varepsilon = 0.4$, $\beta = 0.5$.

What scenario one actually shows

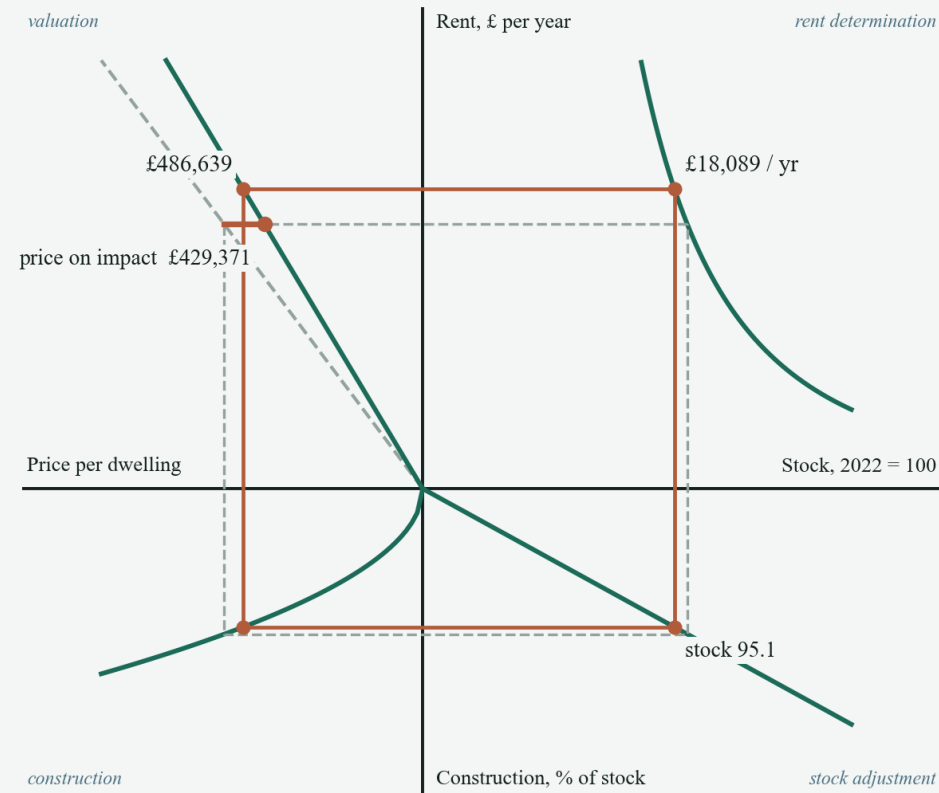
The temptation is to conclude that the model has failed. It has not been tested yet. What has been tested is the assumption that search enquiries measure the demand for space, and that assumption is false in an interesting way. An enquiry is an expression of interest by someone who may already have somewhere to live. The demand the north-east quadrant refers to is the requirement for dwellings, which is driven by household formation and does not fall when people stop browsing.

Between 2022 and 2025 the two moved in opposite directions, and there is a straightforward reason. Mortgage rates roughly trebled, which priced a large number of would-be buyers out of the sales market without removing a single household from the city. They stopped searching for somewhere to buy and carried on needing somewhere to live, which is to say they moved into the rental market. The right lesson is about the mapping from a variable you can observe to a variable the model requires.

Scenario two: dearer capital, and nothing else

F

Leave the demand for space alone and raise the capitalisation rate by seventy-five basis points, from 2.97 to 3.72 per cent. Price falls on impact to £429,371, building falls with it, the stock drifts down to 95.1, and rent rises to £18,088.



Yield shift assumed, not observed. Sold-price coverage in this extract stops in 2022.

Why dearer capital raises rent

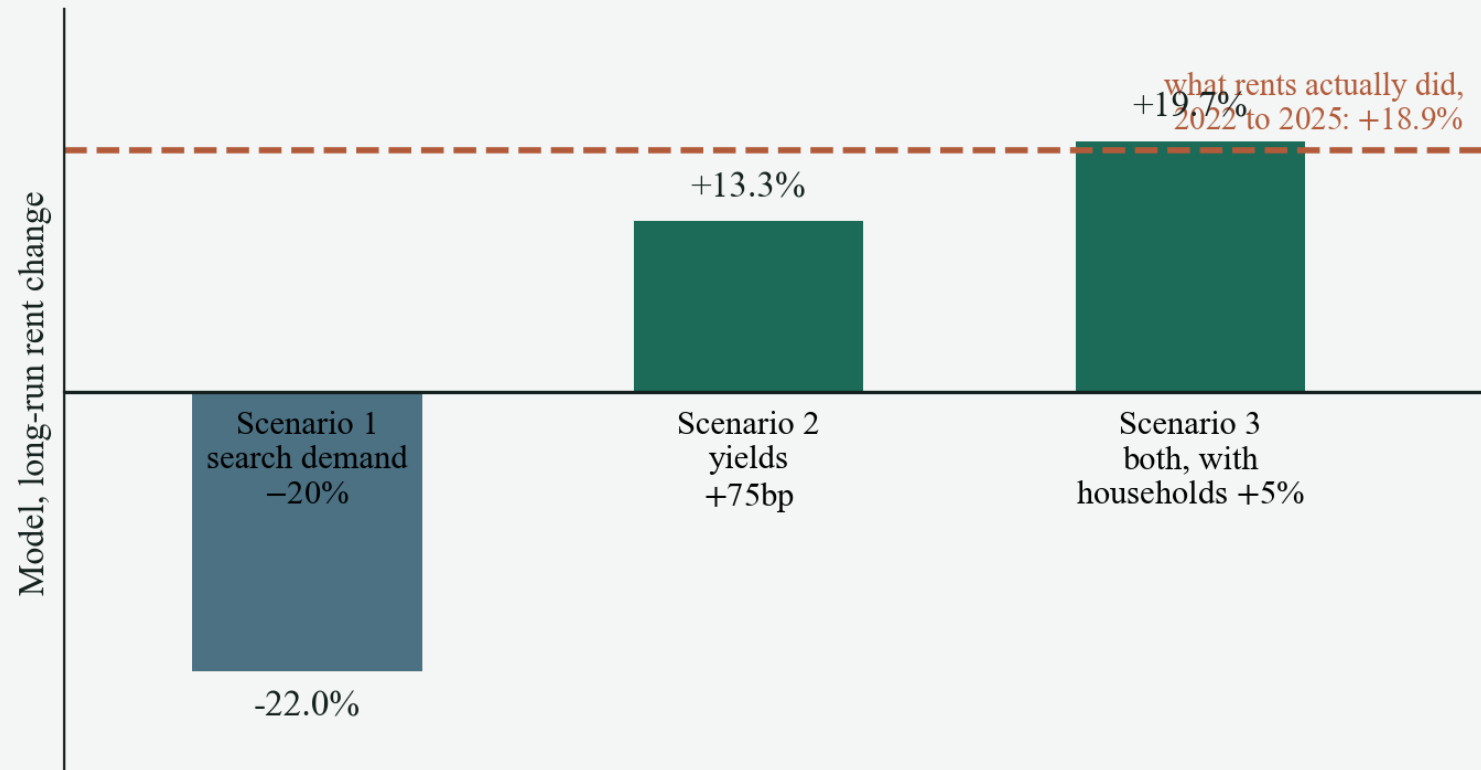
The chain has four steps and none of them involves an occupier changing their mind. A higher required yield lowers what an investor will pay for a given income, so price falls. A lower price makes fewer schemes viable, so building falls. With building below the replacement rate the stock shrinks relative to where it would have been. A smaller stock, facing an unchanged requirement for dwellings, clears at a higher rent.

This is the same result as the yield compression case in Block D, run backwards, and it is the reason that case was worth the time. It also explains something that puzzles people about the period after 2022: interest rates rose, house prices stalled, and rents rose sharply. Those three facts are usually presented as a paradox. In this model they are one event.

Three scenarios against what rents actually did

F

Only the second and third scenarios get the direction right, and the third, which allows both for the rise in yields and for continued household growth of five per cent over three years, lands within a percentage point of the outturn.



Model: $\varepsilon = 0.4$, $\beta = 0.5$. Outturn: advertised rents, Cambridge, 2022 to 2025.

The third scenario is not a validation

Landing within one percentage point of the outturn is not evidence that the model is right, and it is important to say so plainly. Two quantities were chosen freely, the size of the yield rise and the rate of household growth, and neither was observed. Two free parameters fitted to one number will always fit it. Any number of other combinations would have done as well: demand growth of seventeen per cent on its own reproduces the outturn exactly, and so does a yield rise of one hundred and eight basis points on its own.

What the exercise does establish is narrower and more useful. The model tells you what has to have been true for what you observed to make sense, and it rules out a good deal. It rules out the reading in which falling search demand should have pushed rents down. It shows that a yield rise on the scale that plausibly occurred accounts for most of the rent increase without any help from population at all. Used as a decomposition it is informative. Used as a forecast it is not, because its inputs are not forecastable.

How useful is the model, honestly

- It gets the sign right whenever the input is the right variable. Both scenarios that used a genuine driver of the demand for space or the cost of capital pointed in the direction rents actually moved.
- It is unforgiving about inputs. Feed it a plausible-looking series that measures something adjacent, as scenario one did, and it returns a confident answer with the wrong sign.
- It cannot forecast, because it needs future yields and future household growth, which are exactly as hard to know as the rent you were trying to predict.
- It is a good decomposition device. Given an outturn, it says which combinations of shocks are consistent with it and which are not, which is a real service when the public explanation is usually a single cause.
- It says nothing about timing. Three years passed between the base and the outturn, and the model has no view on whether the adjustment took one year or ten.
- Its silence on vacancy matters here. Part of the Cambridge rent rise ran through a fall in available lettings that the model has no way of representing.

Block F discussion

Seminar questions

1. Scenario one used search enquiries as the demand shifter and produced a result with the wrong sign. Name another observable you might have used instead, and say what would be wrong with that one.
2. Demand growth of seventeen per cent alone reproduces the outturn, and so does a yield rise of a hundred and eight basis points alone. What evidence would let you choose between them?
3. The replacement rate changes construction but not rent. Construct a version of the model in which it would change rent.
4. You are asked to forecast Cambridge rents for 2029 using this model. What would you need, and would you take the job?

Block G

What the model leaves out

The standard criticisms, and what they would take to repair.

Indicative length 25 minutes

Colwell on the limits of the diagram

- The model shows two long-run equilibria and the rectangle that connects each to itself. It does not show the path from one to the other, nor how the first was reached.
- The link to the capital market is exogenous and enters through a single number. It would be more natural to treat the interest rate as given from outside and let the capitalisation rate be determined within the model.
- The north-west quadrant relates rent to price, but what is actually capitalised is net operating income. Introducing an operating expense ratio changes the relationship to price equals net income divided by the yield, which is not the same line.
- Several policy questions cannot be posed at all in the diagram as drawn, among them the effect of a change in the treatment of depreciation for tax, or of a cut in the tax on capital gains.
- Volatility in interest rates has no representation. The model has a cap rate but no distribution around it, so it cannot say anything about risk.

What Lisi adds

Lisi takes the criticism that the model has no account of how a market moves between equilibria and supplies one, by putting a search and matching structure inside the quadrants. Space and occupiers do not meet instantly. A landlord holds an asking rent and a prospective tenant holds an offer rent, and the transaction happens when the two converge. Vacancy is then not a failure of the model but an equilibrium outcome of the search process.

The substantive gain is that the framework can now say something about the timing of rent adjustment rather than only about its direction, which is what a forecaster actually needs.

What the model does well, and where it fails

Where it is reliable

- It gets the direction of every first-order comparative static right, which is more than can be said for most intuitive reasoning about the sector.
- It makes the two-market structure explicit and therefore stops students treating rent and yield as the same variable.
- It shows why supply elasticity governs the size of a rent spike, which is the single most useful policy result in real estate economics.
- It is simple enough to be drawn from memory in an examination, and complete enough to be worth drawing.

Where it misleads

- It is silent on timing. Everything happens either instantly or in the long run, and the interesting questions live in between.
- It has no vacancy, so it cannot represent the main variable practitioners actually watch.
- It treats the capitalisation rate as given, which assumes away the interaction that produced the last two cycles.
- It is a single-market model. It cannot represent the differences between submarkets that determine most investment decisions.

Block G discussion

Seminar questions

1. Rewrite the north-west quadrant with an operating expense ratio. What happens to the equilibrium rectangle, and in which direction?
2. If the capitalisation rate were made endogenous, what would determine it, and what new exogenous variable would take its place?
3. The model has no vacancy. Does that make it useless for forecasting, or merely incomplete? Defend your answer.

Block H

Reserve: where the model is used next

What to keep, and what to read.

Indicative length 20 minutes

Where each quadrant reappears

Quadrant	Where it returns	In what form
North-east, rent determination	The sector lectures and the first supervision	Rent as a function of employment and stock, estimated rather than assumed
North-west, valuation	The valuation lectures	The capitalisation rate decomposed into its risk, growth and cost of capital components
South-west, construction	Development appraisal	The threshold price becomes the residual land value
South-east, stock adjustment	Week 8, and any discussion of obsolescence	Depreciation and the question of what happens to space that is no longer wanted

Essential reading

- diPasquale, D. and Wheaton, W. (1992) The markets for real estate assets and space: a conceptual framework. *Real Estate Economics* 20(1), 181 to 197.
- Colwell, P. F. (2002) Tweaking the diPasquale and Wheaton model. *Journal of Housing Economics* 11(1), 24 to 39.

Further reading

- Barras, R. (2009) Building Cycles: Growth and Instability. Oxford: Blackwell-Wiley.
- IPF (2004) Liquidity in Commercial Property Markets. London: Investment Property Forum.
- Jordà, Ò., Schularick, M. and Taylor, A. M. (2015) Leveraged bubbles. *Journal of Monetary Economics* 76, S1 to S20.
- Kuttner, K. N. and Shim, I. (2016) Can non-interest rate policies stabilize housing markets? Evidence from a panel of 57 economies. *Journal of Financial Stability* 26, 31 to 44.
- Lisi, G. (2015) Real estate macroeconomics and the four-quadrant model. *Journal of Real Estate Practice and Education* 18(1), 87 to 106.
- Lisi, G. (2020) Tweaking the diPasquale, Wheaton and Colwell model. *Letters in Spatial and Resource Sciences* 13, 201 to 208.
- Murphy, K. M., Shleifer, A. and Vishny, R. W. (1991) The allocation of talent: implications for growth. *Quarterly Journal of Economics* 106(2), 503 to 530.
- Zhao, S. X., Zhan, H., Jiang, Y. and Pan, W. (2017) How big is China's real estate bubble and why hasn't it burst yet? *Land Use Policy* 64, 153 to 162.

Where this leaves us

Real estate is priced in two markets that move at different speeds, and most of the time they disagree. Disequilibrium is the normal condition of the sector rather than a departure from it.

A large real estate sector is not automatically a productive one. Whether the returns come from producing space or from holding it is an empirical question, and it is the right question to ask of any market you write about.

The four-quadrant model ties the two markets together by requiring that rent, price, construction and stock be mutually consistent. Shift any one schedule and the new equilibrium is the rectangle that closes against it.

The model is reliable about direction and silent about timing. The criticisms of it, particularly Colwell on the treatment of net income and the capitalisation rate, are as much a part of the material as the diagram itself.